RYAN GAO

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SUMMARY

Data specialist with a degree in Astrophysics from UC Berkeley and certificates in Data Science and Full-Stack Development. Experienced in building automated ETL pipelines in Python, evaluating machine learning models, and modeling spatial data. Focuses on empirical analysis, quantitative reasoning, reproducible pipelines, and clear technical documentation.

SKILLS AND TOOLS

- **Languages & Core Tools:** Python (Pandas, NumPy, GeoPandas), SQL+PostgreSQL, JavaScript, HTML/CSS, Git/GitHub
- **Data Operations & Engineering:** Data Annotation, Prompt Testing, Model Auditing, ETL Pipelines, Data Cleaning, Data Visualization, Data Modeling, REST APIs, Docker, MongoDB
- **Quantitative Methods:** Statistical Modeling, Numerical Analysis, Signal Processing, Calculus, Linear Algebra

EXPERIENCE

Data Labeler | Workada (Remote) | 08/2026 – Present

- Evaluate and compare paired model outputs against prompts across specific categories.
- Write qualitative justifications detailing comparative model strengths, flaws, and trade-offs.
- Audit model responses against source materials to identify factual errors, logical flaws, and edge-case failures.

Math Instructor | Elite Education (Hong Kong) | 09/2024 – 10/2025

- Tutored students in algebra, trigonometry, geometry, and calculus to prepare them for standardized testing and upcoming school curriculum
- Wrote feedback after every session detailing the content and material used, and the quality of the student's work.
- Made tailored syllabus, quantitatively tracked student progress, and provided benchmark assessments.
- Worked with the counseling team to steer students towards their goals and adjust their trajectory.

Volunteer Data Analyst | Hack for LA (Los Angeles, CA) | 10/2019 – 02/2020

- Built Python and Pandas ETL pipelines to ingest, clean, and process high-volume municipal 311 service call datasets.
- Contributed to backend service endpoints using Sanic and assisted with containerization using Docker.

Research Intern | Breakthrough Listen / Berkeley SETI (Berkeley, CA) | 05/2016 – 08/2016

- Analyzed radio frequency interference (RFI) patterns by evaluating ground-level radio signals against baseline observation survey data.
- Wrote Python scripts to isolate candidate signals, perform Fast Fourier Transforms (FFT), and produce spectral plots.
- Documented all radio processing work in Jupyter notebooks as reference for future interns and students.

PROJECTS

Redistricting Tracker | Python, GeoPandas, Folium, JavaScript

- Interactive geospatial tool tracking congressional districts altered by mid-decade redistricting
- Ingested district geometries with precinct data and election results to create interactive maps
- Analyzed the impact of redistricting through data modeling, and visualized the results as interactive charts

Associated Press News Scraper | Node.js, React.js, mongoDB, Docker

- Full-stack interactive interface application containerized with Docker
- Programmatically extracts articles from the Associated Press, cleans and stores records into MongoDB

EDUCATION

University of California, Berkeley | Bachelor of Arts in Astrophysics | 2011 – 2016

- Key Coursework: Multivariable Calculus, Linear Algebra, Radio Astronomy Laboratory, Quantum Mechanics

MITx Online | MicroMasters Credential in Data Science | 2018 – 2019

- Key Coursework: Probability & Statistics, Data Analysis for Social Scientists, Machine Learning with Python

UCLA Extension | Certificate in Full-Stack Web Development | 2018 – 2019

- Key Coursework: Database Architecture, Relational SQL, Node.js, React.js

ADDITIONAL

- U.S. Citizen
- Languages: English (Native), Mandarin (Limited Proficiency)